

AERO50001 Aerodynamics 2

Tutorial Sheet 1

Dr G. Papadakis, g.papadakis@ic.ac.uk, Room GAGB 331

Unless otherwise stated, assume standard atmosphere values of $\rho = 1.225 \text{ kg/m}^3$, $\mu = 1.79 \times 10^{-5} \text{ kg/m/s}$, $R = 287.1 \text{ J/(kgK)}$ and $\gamma = 1.4$.

1. A 1750 kg four-wheel drive vehicle is to be delivered to inaccessible country by parachute. What parachute canopy diameter is required to ensure the descent velocity does not exceed 4 m/s, if $C_D = 0.75$ is assumed? You may neglect the mass of the parachute itself.

$$C_D = \frac{D}{\frac{1}{2}\rho U^2 S}$$

$$= \frac{mg}{\frac{1}{2}\rho U^2 S}$$

$$\therefore S = \frac{mg}{\frac{1}{2}\rho U^2 C_D} = \pi \left(\frac{d}{2}\right)^2 = \frac{\pi d^2}{4}$$

$$\therefore d = \left(\frac{4S}{\pi}\right)^{\frac{1}{2}} = \left(\frac{8mg}{\pi\rho U^2 C_D}\right)^{\frac{1}{2}} = 54.53 \text{ m}$$

2. What is the Reynolds number per meter length for an aircraft flying at $M = 0.85$ at sea level?

$$\therefore M = \frac{U}{a}$$

$$\therefore U = Ma$$

$$\begin{aligned}\therefore \frac{Re}{L} &= \frac{\rho U}{\mu} = \frac{\rho Ma}{\mu} = \frac{\rho M \sqrt{\gamma RT}}{\mu} \\ &= \frac{1.225 \times 0.85 \times \sqrt{1.4 \times 287.1 \times 288.2}}{1.79 \times 10^{-5}} \\ &= 1.9798 \times 10^7\end{aligned}$$

3. Wind tunnel tests are to be made on a $\frac{1}{30}$ scale model of an aircraft, to determine the pressure distribution when flying steadily at a speed of 450 m/s and at an altitude of 12 km. The temperature of the air in the working section of the wind tunnel is -73°C and at this temperature $\mu = 1.34 \times 10^{-5} \text{ kg/(ms)}$. (At 12 km, $\mu = 1.42 \times 10^{-5} \text{ kg/(ms)}$, $\rho = 0.311 \text{ kg/m}^3$ and $T = 216.5 \text{ K}$). Determine the absolute static pressure required in the working section for correct simulation of both Reynolds number and Mach number.

$$Re = \frac{\rho U D}{\mu} = \text{constant}$$

$$\therefore \frac{\rho_f U_f D_f}{\mu_f} = \frac{\rho_s U_s D_s}{\mu_s}$$

$$\therefore \rho_s U_s = \rho_f U_f \frac{D_f}{D_s} \frac{\mu_s}{\mu_f}$$

$$\text{and } M = \frac{U}{a} = \frac{U}{\sqrt{\gamma R T}} = \text{constant}$$

$$\therefore \frac{U_f}{\sqrt{\gamma R T_f}} = \frac{U_s}{\sqrt{\gamma R T_s}}$$

$$\therefore U_s = U_f \sqrt{\frac{T_s}{T_f}}$$

$$\therefore \rho_s = \rho_f \frac{D_f}{D_s} \frac{\mu_s}{\mu_f} \sqrt{\frac{T_f}{T_s}}$$

$$\therefore P_s = \rho_s R T_s = \rho_f \frac{D_f}{D_s} \frac{\mu_s}{\mu_f} R \sqrt{T_f T_s}$$

$$= 5.26 \times 10^5 \text{ Pa}$$

$$\therefore |P_s| = 5.26 \times 10^5 \text{ Pa}$$

4. Wind tunnel tests are to be carried out to estimate the wind loading acting on a building. Dynamic similarity can be achieved by using a $\frac{1}{30}$ scale model in a compressed air tunnel with a working section operating at 25 atmospheres and atmospheric temperature. Explain why there is no need to match Mach number. If the atmospheric wind velocity is 15 m/s, what is the air speed needed in the tunnel? Under these conditions what will be the ratio of the drag of the model to that of the full scale building?

Because in this case $M = \frac{U}{a} = \frac{U}{\sqrt{\gamma RT}} = 0.0441 < 0.3$,

the effect of compressibility is unlikely to be important.

$$\therefore Re = \frac{\rho U D}{\mu} = \text{constant}$$

$$\therefore \mu \propto T$$

$$\therefore \frac{\rho_f U_f D_f}{T_f} = \frac{\rho_s U_s D_s}{T_s} \quad \text{while } T_f = T_s$$

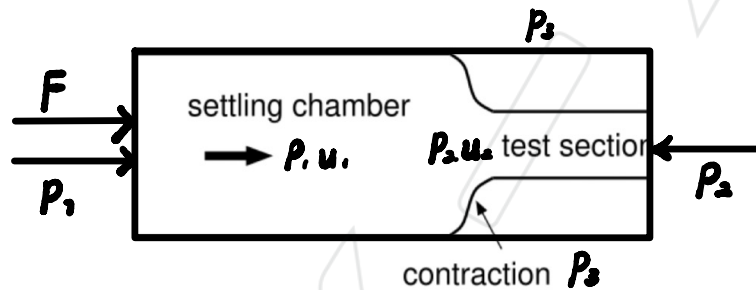
$$\therefore P = \rho RT \quad \text{and} \quad P_s = 25 P_f \quad \therefore \rho_s R T_s = 25 \rho_f R T_f$$

$$\therefore \rho_s = 25 \rho_f \frac{R T_f}{R T_s} = 25 \rho_f$$

$$\therefore U_s = U_f \frac{\rho_f}{\rho_s} \frac{D_f}{D_s} \frac{T_s}{T_f} = U_f \frac{\rho_f}{\rho_s} \frac{D_f}{D_s} = 18 \text{ m/s}$$

$$\therefore \frac{D_s}{D_f} = \frac{\frac{1}{2} \rho_s U_s^2 A_s C_D}{\frac{1}{2} \rho_f U_f^2 A_f C_D} = \frac{\rho_s}{\rho_f} \left(\frac{U_s}{U_f} \right)^2 \left(\frac{D_s}{D_f} \right)^2 = \frac{1}{25}$$

5. The settling chamber, contraction and test section of a wind tunnel are shown below. The settling chamber and test section are both square with dimensions $2.743\text{m} \times 2.743\text{m}$ and $0.9144\text{m} \times 0.9144\text{m}$ respectively (this is our so-called 3ft by 3ft tunnel). The internal pressure of the test section is maintained at atmospheric. The walls of the settling chamber are constructed from $2.743\text{m} \times 1.219\text{m}$ wooden panels. When the test section velocity is 32m/s :
- (a) what is the force exerted on the settling chamber panels due to the internal pressure? Assume that the flow between the settling chamber and the test section can be treated as loss free.
- (b) what is the magnitude and direction of the force acting on the contraction of the wind tunnel?



(a) Assume the flow is steady and incompressible

$$\therefore \dot{m} = \rho u A = \text{constant}$$

$$\therefore u_1 A_1 = u_2 A_2$$

$$\therefore u_1 = u_2 \frac{A_2}{A_1} = 3.556 \text{ m/s}$$

$$\therefore P_1 = P_1 + \frac{1}{2} \rho u_1^2 = P_2 + \frac{1}{2} \rho u_2^2 = P_{\text{atm}} + \frac{1}{2} \rho u_2^2$$

$$\therefore \Delta P = P_1 - P_{\text{atm}} = \frac{1}{2} \rho (u_2^2 - u_1^2) = 619.45 \text{ Pa}$$

$$\therefore F_3 = \Delta P A_3 = 2071.28 \text{ N}$$

b) According to the conservation of momentum, set CV as shown above, for single-dimension flow:

$$\rho(u_2^2 A_2 - u_1^2 A_1) = F + P_1 A_1 - P_2 A_2 - P_3 (A_1 - A_2)$$

$$\therefore P_1 + \frac{1}{2}\rho u_1^2 = P_2 + \frac{1}{2}\rho u_2^2 = P_3 + \frac{1}{2}\rho u_2^2$$

$$\therefore P_3 = P_2$$

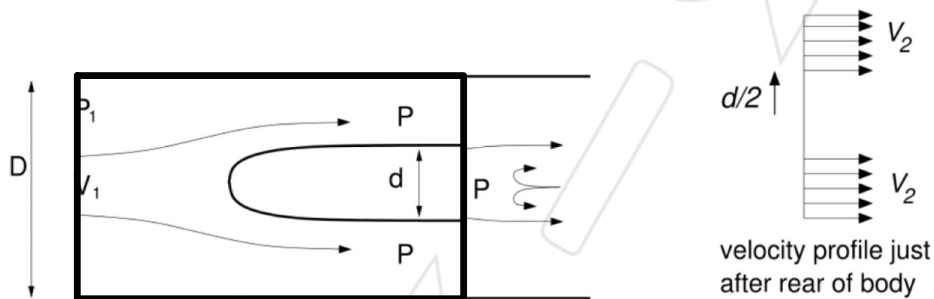
$$\therefore \rho_1 u_1 A_1 = \rho_2 u_2 A_2$$

$$\therefore u_1 = u_2 \frac{A_2}{A_1} \quad \therefore P_1 - P_2 = \frac{1}{2}\rho(u_2^2 - u_1^2) = \frac{1}{2}\rho u_2^2 \left(1 - \frac{A_2^2}{A_1^2}\right)$$

$$\begin{aligned}\therefore F &= \rho u_2^2 A_2 \left(1 - \frac{A_2^2}{A_1^2}\right) - (P_1 - P_2) A_1 \\ &= \rho u_2^2 A_2 \left(1 - \frac{A_2^2}{A_1^2}\right) - \frac{1}{2}\rho u_2^2 A_1 \left(1 - \frac{A_2^2}{A_1^2}\right) \\ &= -3748.52 \text{ N}\end{aligned}$$

$$\therefore F = 3748.52 \text{ N from right to left}$$

6. Consider incompressible flow over a cylindrical body, diameter d , placed centrally in a pipe of diameter D , as shown below. The body has a streamlined nose but the rear is cut off square. The pressure P over the rear of the body is equal to that in the mainstream flow at the point of separation (where there is no streamline curvature). The velocity profile just after the rear of the body can be assumed to have the form shown below.
- Use continuity to express V_2 in terms of V_1 , D and d .
 - Obtain an expression for P in terms of ρ , P_1 , V_1 and V_2 , clearly stating any assumptions made. [Hint: Bernoulli can be used between locations **upstream** of any losses.]
 - Use a control volume momentum analysis to obtain an expression for the drag coefficient of the body in terms of d/D . Neglect boundary layer growth and assume incompressible flow.



(i) According to the continuity equation,

$$\rho V_1 A_1 = \rho V_2 A_2 = \text{constant for steady flow}$$

$$\therefore \rho V_1 \pi \frac{D^2}{4} = \rho V_2 \pi \left(\frac{D^2}{4} - \frac{d^2}{4} \right)$$

$$\therefore V_2 = \frac{D^2}{D^2 - d^2} V_1$$

(ii) At upstream of losses, for steady flow

$$P_t = P_1 + \frac{1}{2} \rho V_1^2 = P + \frac{1}{2} \rho V_2^2$$

$$\therefore P = P_1 + \frac{1}{2} \rho (V_1^2 - V_2^2)$$

ciii) for incompressible - steady flow,

according to the conservation of momentum in CV

$$\rho (u_2^2 A_2 - u_1^2 A_1) = F + P_1 A_1 - P_2 A_2 - P_3 (A_1 - A_2)$$

$$\therefore P_2 = P_3 = P \quad \therefore -P_2 A_2 - P_3 (A_1 - A_2) = -P A_1$$

$$\therefore \rho (V_2^2 A_2 - V_1^2 A_1) = D + (P_1 - P) A_1$$

$$\therefore D = \rho (V_2^2 A_2 - V_1^2 A_1) + \frac{1}{2} \rho (V_1^2 A_1 - V_2^2 A_1)$$

$$= \rho (V_2^2 A_2 - \frac{1}{2} V_1^2 A_1 - \frac{1}{2} V_2^2 A_1)$$

$$\therefore \rho V_1 A_1 = \rho V_2 A_2 \quad \therefore V_2 A_2 = V_1 A_1 \quad \therefore V_2 = V_1 \frac{A_1}{A_2}$$

$$\therefore D = \rho (V_1 V_2 A_1 - \frac{1}{2} V_1^2 A_1 - \frac{1}{2} V_2^2 A_1)$$

$$= \rho A_1 (V_1 V_2 - \frac{1}{2} V_1^2 - \frac{1}{2} V_2^2)$$

$$= -\frac{1}{2} \rho A_1 (V_1^2 - 2V_1 V_2 + V_2^2) = -\frac{1}{2} \rho A_1 (V_1 - V_2)^2$$

$$= -\frac{1}{2} \rho A_1 (V_1 - \frac{D^2}{D^2 - d^2} V_1)^2 = \frac{1}{2} \rho A_1 (\frac{d^2}{D^2 - d^2} V_1)^2$$

$$\therefore C_o = \frac{D}{\frac{1}{2}\rho V_i^2 A_i}$$

$$= \frac{d^4}{(D^4 - d^4)^2}$$

$$= \frac{\left(\frac{d}{D}\right)^4}{\left(1 - \left(\frac{d}{D}\right)^4\right)^2}$$

7. A fluid of density ρ is rotating as a solid body (forced vortex) about an axis at Ω rad/s. Calculate the total pressure $P_o(r)$ at radius r relative to the static pressure p_s at the axis.

For forced vortex,

$$V_\theta = CR = \Omega R$$

$$V_r = 0$$

$$\therefore V = \Omega R$$

$$\therefore \underbrace{p_{s-\text{axis}} + 0}_{\text{at axis}} = \underbrace{p_s + \frac{1}{2}\rho\Omega^2 r^2}_{\text{at } r}$$

$$\begin{aligned}\therefore P_o(r) &= p_{s-\text{axis}} + \frac{1}{2}\rho\Omega^2 r^2 \\ &= p_s + \rho\Omega^2 r^2\end{aligned}$$

Numerical answers:

1. 54.5 m

2. 1.98×10^7

3. 5.25×10^5 Pa

4. 18 m/s, 1:25

5. (a) 2071 N (b) 3730 N

6. $C_D = \frac{(\frac{d}{D})^2}{(1-(\frac{d}{D})^2)^2}$

7. $p_s + \rho\Omega^2 r^2$